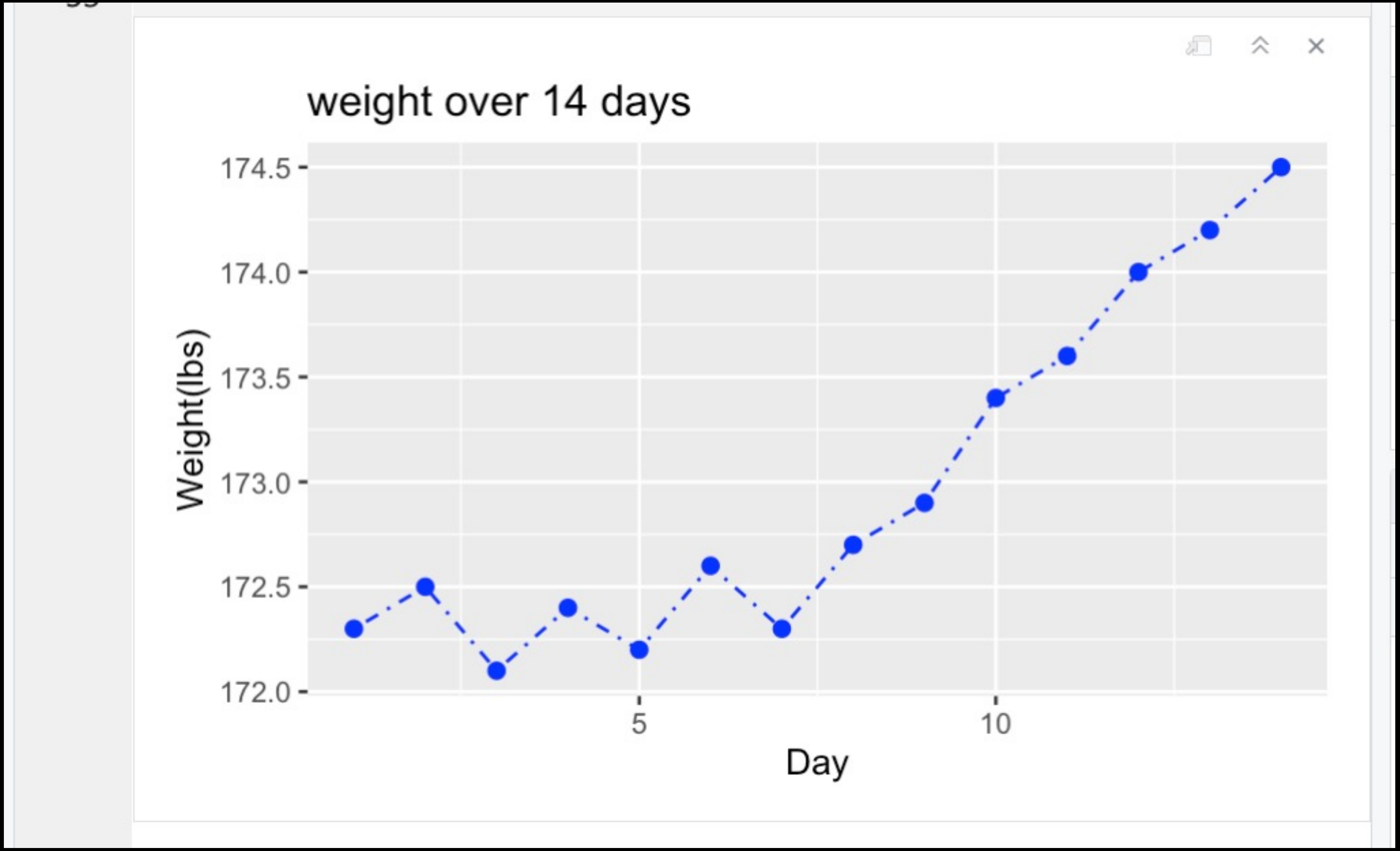
CALORIE INTAKE AND WEIGHT GAIN ANALYSIS

- DERRICK NYAGESUKA DATA211 05/01/2025

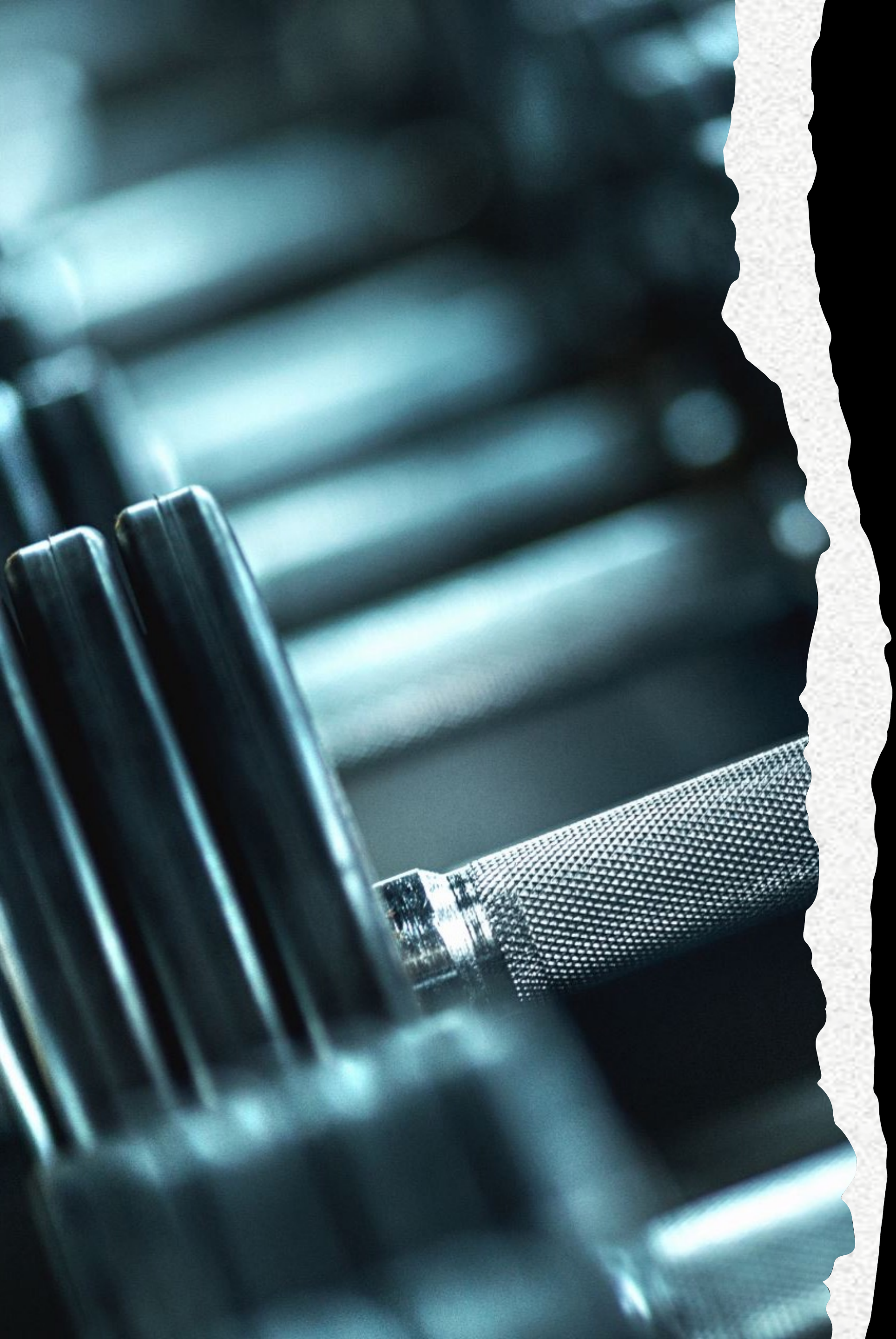
	A	B
1	weight	calories
2	172.3	2500
3	172.5	2500
4	172.1	2500
5	172.4	2500
6	172.2	2500
7	172.6	2500
8	172.3	2500
9	172.7	3500
10	172.9	3500
11	173.4	3500
12	173.6	3500
13	174	3500
14	174.2	3500
15	174.5	3500
16		
17		

Question asked:

- Does increasing calorie intake from 2500 to 3500 calories increase weight gain?
- The body weight was the dependent variable.
- Calorie intake was the independent variable.



Line graph showing weight gain trend



Hypothesis testing

- The question asked in this analysis is, does increasing my daily calorie intake from 2500 to 3500 calories for 7 days result in greater weight gain compared to maintaining 2500 calories for 7 days?
- I chose the question as I needed to understand how calorie intake affects weight which is crucial for health and fitness goals and thought that a 1000 calorie surplus would be efficient to note any significant short term weight gain.

Null Hypothesis:

H_0: μ_Increased = μ_Normal

Alternative Hypothesis:

H_1: μ_Increased > μ_Normal

```
        Welch Two Sample t-test

data:  increasedWeight and normalWeight
t = 4.8845, df = 6.7931, p-value = 0.0009715
alternative hypothesis: true difference in means is greater than
0
95 percent confidence interval:
 0.7760039       Inf
sample estimates:
mean of x mean of y
 173.6143  172.3429
```

Conclusion

- With a p-value of $0.0009715<0.05$, at 5% significance level, I have enough evidence to reject the null hypothesis and conclude that increasing my calorie intake from 2500 calories to 3500 calories leads to greater weight gain.